Square Root QR Covariance Steering

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Abstract

This document provides an overview and technical documentation for the Square Root QR Covariance Steering method, including problem formulation, algorithmic details, and implementation notes.

1 Introduction

Covariance steering is a control technique for steering the mean and covariance of a stochastic system to desired terminal values. The proposed square root QR approach improves numerical stability and efficiency, at the expense of an iterative algorithm.

1.1 Notation

Let $\mathbb{L}_+^n$ denote the space of $n \times n$ lower triangular matrices with positive diagonal entries. The square root of a positive definite matrix $P \in \mathbb{R}^{n\times n}$ is defined as the unique lower triangular matrix $S \in \mathbb{L}_+^n$ such that $P = SS^\top$. Its uniqueness is easily shown. $\operatorname{diag}(M)$ denotes the vector of diagonal elements of matrix M. The linear matrix inequality $A \succ (\succeq) B$ means that $A - B$ is positive (semi)definite. $a\colon b$ denotes the set of integers $a, a+1, \dots, b$.

1.2 Problem Formulation

We address the finite-horizon discrete-time covariance steering problem for a linear time-varying system with additive Gaussian noise. The goal is to find a state feedback controller that steers the state covariance from an initial value to a desired terminal value while minimizing a cost function, and under chance constraints. Mathematically, the problem is formulated as follows.

$$
\begin{aligned}
\underset{\{v_k, K_k\}_{k=0}^{N-1}}{\text{minimize}} \quad & J = \sum_{k=0}^{N-1} J_k(x_k, u_k) && \text{(1a)}\\
\text{subject to} \quad & x_{k+1} = A_k x_k + B_k u_k + G_k w_k, \quad k \in 0\colon N-1 && \text{(1b)}\\
& u_k = v_k + K_k(x_k - \mathbb{E}[x_k]), \quad k \in 0\colon N-1 && \text{(1c)}\\
& x_0 \sim \mathcal{N}(\mu_{\text{init}}, P_{\text{init}}) && \text{(1d)}\\
& x_N \sim \mathcal{N}(\mu_{\text{fin}}, P_{\text{fin}}), \quad P_{\text{fin}} \succeq \mathbb{E}\left[(x_N - \mathbb{E}[x_N])(x_N - \mathbb{E}[x_N])^\top\right] && \text{(1e)}\\
& \mathbb{P}(x_k \in \mathcal{X}_k) \geq 1 - \epsilon_k, \quad k \in 0\colon N && \text{(1f)}\\
& \mathbb{P}(u_k \in \mathcal{U}_k) \geq 1 - \delta_k, \quad k \in 0\colon N-1 && \text{(1g)}
\end{aligned}
$$

where $x_k \in \mathbb{R}^n$ is the state vector, $u_k \in \mathbb{R}^m$ is the control input, $w_k \in \mathbb{R}^n$ is standard Gaussian process noise, $v_k \in \mathbb{R}^m$ is the nominal control input, and $K_k \in \mathbb{R}^{m\times n}$ is the state feedback gain to be designed. The state-affine feedback was shown to be optimal for the covariance steering problem without chance constraints [?]. The matrices A_k, B_k, and G_k define the system dynamics. We consider two types of cost functions:

$$
J_k(x_k, u_k) = \begin{cases} \mathbb{E}\left[x_k^\top Q_k x_k + u_k^\top R_k u_k\right], \quad Q_k \succeq 0, R_k \succ 0 & \text{(quadratic cost)} \\ \mathcal{Q}_{\|u_k\|_2}(p_J) & \text{(quantile 2-norm cost)} \end{cases} \tag{2}
$$

where $\mathcal{Q}_{\|u_k\|_2}(p_J)$ is p_J-quantile of the 2-norm of the control input u_k.

The sets $\mathcal{X}_k$ and $\mathcal{U}_k$ represent state and control constraints, respectively, with associated violation probabilities ϵ_k and δ_k. We assume that the feasible region defined by the chance constraints is given in one of two forms: a linear constraint or a norm constraint. For the linear constraint form, the chance constraint can be expressed as:

$$\mathbb{P}(\alpha_j^\top x_k \leq \beta_j) \geq 1 - p_j \tag{3}$$

for each linear constraint defined by $\alpha_j \in \mathbb{R}^n$ and $\beta_j \in \mathbb{R}$, with violation probability p_j. For the norm constraint form, the chance constraint can be expressed as:

$$\mathbb{P}(\|x_k\|_2 \leq \gamma) \geq 1 - p_\gamma \tag{4}$$

for some scalar $\gamma > 0$ and violation probability p_γ. The same forms apply to the control input u_k.

For convenience, we make the following assumption, which is reasonable in practical applications:

Assumption 1. *The state covariance remains strictly positive definite throughout the horizon.*

1.3 Relevant works

The covariance steering problem has been well studied in the literature, and dates back to the early works by Hotz and Skelton [2]. Recent works mainly focus on converting the problem into a convex optimization problem. Skaf and Boyd [13] considers a state-history feedback controller and formulates the problem as a semidefinite programming (SDP) problem. Okamoto et al. [12] later adds chance constraints while maintaining convexity. However, these methods suffer from the poor computational complexity as the horizon length increases, due to the large linear matrix inequality (LMI) constraints involved [**?**]. It is also important to note that the problem variables involve the "square root" of the covariance.

A more recently proposed method by Liu et al. [**?**] uses the covariance directly as the optimization variable; in combination with an elegant proof of so-called "lossless convexification", it converts the seemingly nonconvex problem into an equivalent convex problem. Compared to the block approach, this method scales much better with the growth in problem size. At the same time, we have observed several drawbacks of this method: first, the numerical stability is not ideal as we are working with units that square the system units. Second, when dealing with objective functions or constraints that involve the covariance square root (e.g., the chance constraints), iterative algorithms are required. Third, due to the requirement of a proof of lossless convexification, whenever the problem formulation changes slightly, theoretical work is required to ensure the convexity of the new formulation.

1.4 Contributions

2 Preliminaries

2.1 QR decomposition

Lemma 1 (Existence and uniqueness of economy-size QR decomposition, p.75 of [3])**.** *Let $M \in \mathbb{R}^{m\times n}$, $m > n$.*

- *(Existence) There exist $Q \in \mathbb{R}^{m\times n}$ and $R \in \mathbb{R}^{n\times n}$ such that $QQ^\top = I$ and R is upper triangular, and $M = QR$.*
- *(Uniqueness) If M has full rank, i.e.* $\operatorname{rank}(M) = n$*, then the matrices Q and R are unique up to a sign change of the diagonal entries of R.*

Remark 1. *If we require the diagonal entries of R to be positive, then the sign ambiguity is resolved. R is invertible.*

2.2 Square root covariance propagation

First, we provide a brief review of square root covariance propagation, mainly used for Kalman filtering [14]. Let the system dynamics be given by:

$$x_{k+1} = \Phi_k x_k + G_k w_k \tag{5}$$

with state $x_k \in \mathbb{R}^n$, where $w_k \in \mathbb{R}^m$ is standard Gaussian noise. By defining $P_k := \mathbb{E}\left[(x_k - \mathbb{E}[x_k])(x_k - \mathbb{E}[x_k])^\top\right]$, the covariance propagation equation is given by:

$$P_{k+1} = \Phi_k P_k \Phi_k^T + G_k G_k^\top \tag{6}$$

If we define $S_k \in \mathbb{L}_+^n$ as $S_k = P_k^{1/2}$, the covariance propagation can be expressed as:

$$S_{k+1} S_{k+1}^T = \Phi_k S_k S_k^T \Phi_k^T + G_k G_k^\top = X_{k+1} X_{k+1}^T \tag{7}$$

where $S_{k+1} \in \mathbb{L}_+^n$ is the next step square root covariance, and we define

$$X_{k+1} := \begin{bmatrix} \Phi_k S_k & G_k \end{bmatrix} \in \mathbb{R}^{n \times (n+m)} \tag{8}$$

Note that the dimensions of S_{k+1} and X_{k+1} are different, so we cannot directly equate them. For a matrix $Q_k \in \mathbb{R}^{(n+m)\times n}$ such that $Q_k Q_k^\top = I$, (is this iff?)

$$Q_k S_{k+1}^\top = X_{k+1}^\top \implies X_{k+1} X_{k+1}^\top = S_{k+1} Q_k^\top Q_k S_{k+1}^\top = S_{k+1} S_{k+1}^\top. \tag{9}$$

Since $S_{k+1} \in \mathbb{L}_+^n$, its transpose is upper triangular, so the first equality in (9) is an economy-size QR decomposition of $X_{k+1}^\top$. By Lemma 1, if $X_{k+1}^\top$ is full rank, there exists a unique set of matrices Q_k and S_{k+1} that satisfy the equation. In other words, the square root covariance propagation can be expressed as a nonlinear operation:

$$S_{k+1} = (\mathrm{qr}(X_{k+1}^\top))^\top = (\mathrm{qr}(\begin{bmatrix} \Phi_k S_k & G_k \end{bmatrix}^\top))^\top \tag{10}$$

where $\mathrm{qr}(\cdot)$ denotes the operation of extracting the upper triangular factor R from the economy-size QR decomposition. [1]

2.3 QR decomposition derivative

In order to use this square root covariance propagation in an optimization framework, we require the derivative of the QR decomposition operation; this is documented in the literature [1], but for completeness, we provide a brief overview here. The derivative of the QR decomposition, i.e. the first-order perturbation analysis of the R matrix with respect to perturbations in the X matrix, is given by:

$$A + (\mathrm{d}R)R^{-1} = Q^\top(\mathrm{d}X)R^{-1} \tag{11}$$

where A is an antisymmetric (skew-symmetric) matrix such that

$$\mathrm{tril}(A) = \mathrm{tril}(Q^\top(\mathrm{d}X)R^{-1}). \tag{12}$$

$\mathrm{tril}(A)$ replaces the upper triangular part of A with zeros. Organizing the above, we have:

$$\mathrm{d}R = \mathrm{d}R(\mathrm{d}X, Q, R) := Q^\top(\mathrm{d}X) - \mathrm{tril}(Q^\top(\mathrm{d}X)R^{-1})R. \tag{13}$$

Thus, using the Taylor expansion, we can approximate the QR decomposition as a linear function of perturbations in X, around some reference $\bar{X} = \bar{Q}\bar{R}$:

$$\mathrm{qr}(X) \approx \mathrm{qr}(\bar{X}) + \mathrm{d}R(X - \bar{X}, \bar{Q}, \bar{R}). \tag{14}$$

Note that $\mathrm{qr}(\bar{X})$ and $\bar{R}$ are synonymous.

[1] For example, in MATLAB, this requires using $\mathrm{qr}(X_{k+1}^\top$, "econ"). In addition, to ensure uniqueness of the square root, if the function returns an R matrix with negative diagonal entries, we multiply the corresponding columns of Q and rows of R by -1 to ensure positive diagonals.

3 Application to Covariance Steering

The application to covariance steering is straightforward.

3.1 Propagation with feedback control

Now, since the state is controlled by a the affine feedback control law, the closed-loop dynamics become

$$x_{k+1} = A_k x_k + B_k(v_k + K_k(x_k - \mathbb{E}[x_k])) + G_k w_k \quad k \in 0\colon N-1. \tag{15}$$

The mean dynamics are given by

$$\mu_{k+1} = A_k \mu_k + B_k v_k, \quad k \in 0\colon N-1. \tag{16}$$

where $\mu_k := \mathbb{E}[x_k]$. The square root covariance dynamics are given by:

$$S_{k+1} = (\mathrm{qr}(X_{k+1}^\top))^\top, \quad X_{k+1} = \begin{bmatrix}(A_k + B_k K_k)S_k & G_k\end{bmatrix} \quad k \in 0\colon N-1. \tag{17}$$

Note that due to the affine feedback control law, the Gaussianity of the state distribution is preserved, and the control input is also Gaussian. Since both P_k and K_k are unknowns, there is a bilinear term $K_k S_k$. Here, we introduce a new variable $L_k = K_k S_k \in \mathbb{R}^{m\times n}$, so that the bilinear term is replaced with a linear term. K_k can be uniquely recovered as $K_k = L_k S_k^{-1}$. The inverse exists when the diagonals of S_k are non-zero (source?), which is by definition guaranteed since S_k is the Cholesky factor of a positive definite covariance matrix ($\because$ Assumption 1). So we get

$$S_{k+1} = (\mathrm{qr}(X_{k+1}^\top))^\top, \quad X_{k+1} = \begin{bmatrix}A_k S_k + B_k L_k & G_k\end{bmatrix} \quad k \in 0\colon N-1. \tag{18}$$

with its first-order approximation given by:

$$S_{k+1} \approx (\mathrm{qr}(\bar{X}_{k+1}^\top))^\top + \left(\mathrm{d}R(X_{k+1}^\top - \bar{X}_{k+1}^\top, \bar{Q}_{k+1}, \bar{R}_{k+1})\right)^\top \quad k \in 0\colon N-1. \tag{19}$$

where $\bar{X}_{k+1} = \begin{bmatrix}A_k \bar{S}_k + B_k \bar{L}_k & G_k\end{bmatrix}$, $\bar{S}_k$ and $\bar{L}_k$ being the reference variables, and $\bar{Q}_{k+1}\bar{R}_{k+1} = \bar{X}_{k+1}$ being the QR decomposition of $\bar{X}_{k+1}$. X is full rank if $AS + BL$ is full rank. If $L = 0$ this is guaranteed. When L is updated at subsequent iterations, what happens?

3.2 Objective function

After substituting $u_k = K_k x_k = L_k S_k^{-1} x_k$, and some standard manipulations, the objective function can be expressed as deterministic convex functions of L_k and S_k:

$$J = \begin{cases} \sum_{k=0}^{N-1} \mu_k^\top Q_k \mu_k + v_k^\top R_k v_k + \mathrm{trace}(Q_k S_k S_k^\top + R_k L_k L_k^\top) & \text{(quadratic cost)} \\ \sum_{k=0}^{N-1} \|v_k\|_2 + \sqrt{\mathcal{Q}_{\chi_m^2}(p_J)}\, \|L_k\|_2 & \text{(quantile 2-norm cost).} \end{cases} \tag{20}$$

where the 2-norm cost is conservatively approximated [11]. $\mathcal{Q}_{\chi_m^2}(\cdot)$ is the inverse of the chi-squared cumulative distribution function with m degrees of freedom.

3.3 Constraints

The initial distributional constraint is imposed by:

$$\mu_0 = \mu_{\mathrm{init}}, \quad S_0 = P_{\mathrm{init}}^{1/2} \tag{21}$$

The terminal distributional constraint is imposed by [12]:

$$\mu_N = \mu_{\mathrm{fin}}, \quad \left\| P_{\mathrm{fin}}^{-1/2} S_N \right\|_2 - 1 \le 0 \tag{22}$$

where $\|\cdot\|_2$ denotes the spectral norm. To ensure uniqueness of the square root, we impose:

$$\mathrm{diag}(S_k) \geq 0, \quad k \in 0\colon N \tag{23}$$

Since we are working with the square root form, chance constraints can also be imposed in a convex manner. For the linear constraint form (3), the chance constraint can be exactly expressed as:

$$\alpha_j^\top \mu_k + \mathcal{Q}_{\mathcal{N}}(1-p_j)\left\|S_k^\top \alpha_j\right\|_2 - \beta_j \leq 0 \quad k \in 0\colon N \tag{24}$$

where $\mathcal{Q}_{\mathcal{N}}(p_j)$ is the inverse of the standard normal cumulative distribution function. For the norm constraint form (4), the chance constraint can be conservatively approximated as:

$$\|\mu_k\|_2 + \sqrt{Q_{\chi_n^2}(1-p_\gamma)}\ \|S_k\|_2 - \gamma \leq 0 \quad k \in 0\colon N \tag{25}$$

The control chance constraints can be imposed by replacing μ_k with v_k and S_k with L_k.

To summarize, (1) is converted into a nonconvex optimization problem as follows:

$$\underset{\{v_k, L_k\}_{k=0}^{N-1}, \{\mu_k, S_k\}_{k=0}^{N}}{\mathrm{minimize}} \ (20) \quad \text{subject to} \quad (18), (21), (22), (23), (24), (25) \tag{26}$$

4 Algorithm Overview

In order to take advantage of the convex representation of the objective and constraints, we use sequential convex programming (SCP). At each iteration, (19) is imposed in a vectorized form with penalization of violations:

$$\mathrm{vectril}\left\{S_{k+1} - (\mathrm{qr}(\bar{X}_{k+1}^\top))^\top - \left(\mathrm{d}R(X_{k+1} - \bar{X}_{k+1}, \bar{Q}_{k+1}, \bar{R}_{k+1})\right)^\top\right\} = \chi_k \tag{27}$$

where $\mathrm{vectril}(\cdot)$ vectorizes the lower triangular part of a matrix, and $\chi_k \in \mathbb{R}^{\frac{n(n-1)}{2}}$ is a slack variable, penalized in the objective function via an Augmented Lagrangian method [10]. By using the specific SCP algorithm `SCvx*`, we can ensure that 1. the nonlinear dynamics are obeyed at convergence, and 2. the converged solution is locally optimal.

To ensure that the solution does not cause artificial infeasibility, we impose a trust region constraint

$$\mathrm{vectril}(S_k - \bar{S}_k) \leq r \quad k \in 0\colon N \tag{28}$$

where r is the trust region radius.

The objective function is augmented with a penalty term as follows:

$$J_{\mathrm{aug}} = J + \sum_{k=0}^{N-1}(\lambda_k^\top \chi_k + \frac{w}{2}\|\chi_k\|_2^2) \tag{29}$$

where λ_k are Lagrange multipliers, and w is the penalty weight.

The subproblem is then given by:

$$\mathrm{minimize}\,(29) \quad \text{subject to} \quad (27), (21), (22), (23), (24), (25), (28) \tag{30}$$

The complete algorithm is summarized in Algorithm 1. The algorithm iteratively solves a sequence of convex optimization problems, where at each iteration, the nonlinear square root covariance dynamics (18) are linearized around the current reference variables. The convex subproblem includes all deterministic constraints (boundary conditions, chance constraints) and the linearized dynamics with slack variables penalized via the Augmented Lagrangian method. Readers are referred to [10] for more details of the `SCvx*` algorithm.

Algorithm 1 Square Root QR Covariance Steering via SCvx*

Require: Initial guess $(\bar{S}_k)_{k=0}^{N}$, $(\bar{L}_k)_{k=0}^{N-1}$
Require: SCP parameters $\epsilon_{\text{feas}} > 0, \epsilon_{\text{opt}} > 0$ among others.
Initialize SCP parameters
Compute the reference QR matrices: $\{\bar{Q}_{k+1}, \bar{R}_{k+1}\} \leftarrow \text{qr}(\begin{bmatrix} A_k\bar{S}_k + B_k\bar{L}_k & G_k \end{bmatrix})$
while $|\Delta J| > \epsilon_{\text{opt}}$ or $\chi > \epsilon_{\text{feas}}$ **do**
  $\{(S_k^*, L_k^*, \mu_k^*, v_k^*), (\chi_k^*)\} \leftarrow \arg\min$ (30)
  if step acceptance conditions are met **then**
    $\bar{S}_k \leftarrow S_k^*, \bar{L}_k \leftarrow L_k^*, \bar{\mu}_k \leftarrow \mu_k^*, \bar{v}_k \leftarrow v_k^*$
    Update the reference QR matrices: $\{\bar{Q}_{k+1}, \bar{R}_{k+1}\} \leftarrow \text{qr}(\begin{bmatrix} A_k\bar{S}_k + B_k\bar{L}_k & G_k \end{bmatrix})$
    Update Lagrange multipliers λ_k and stationarity tolerance
  end if
  Update trust region radius r and penalty weight w
end while
Recover feedback gains: $K_k^* \leftarrow L_k^*(S_k^*)^{-1}$ for $k \in 0\colon N-1$
return $((S_k^*)_{k=0}^{N}, (\mu_k^*)_{k=0}^{N}, (v_k^*)_{k=0}^{N-1}, (K_k^*)_{k=0}^{N-1})$

5 Global Optimality of Converged Solution in the Absence of Chance Constraints.

In this section, we provide a theoretical result regarding the global optimality of solutions to convex optimization problems that are related through a diffeomorphism. Consider two optimization problems:

$$\text{(A)} \quad \underset{x}{\text{minimize}} \quad f(x) \quad \text{s.t.} \quad x \in \mathcal{F}_A \tag{31}$$

$$\text{(B)} \quad \underset{y}{\text{minimize}} \quad g(y) = f(T(y)) \quad \text{s.t.} \quad y \in \mathcal{F}_B \tag{32}$$

where $T : \mathcal{F}_B \to \mathcal{F}_A$ is a diffeomorphism such that $T(\mathcal{F}_B) = \mathcal{F}_A$. (A) is convex, i.e. $f(x)$ is a convex function and $\mathcal{F}_A$ is a convex set. (B) is nonconvex, i.e. without loss of generality, $\mathcal{F}_B$ is a nonconvex set.

Proposition 1 (Global Optimality via Diffeomorphism)**.** *Let $T : \mathcal{F}_B \to \mathcal{F}_A$ be a diffeomorphism such that $T(\mathcal{F}_B) = \mathcal{F}_A$, and let y^* be a local minimum of (B). Then $x^* = T(y^*)$ is a global minimum of (A).*

Proof. Since y^* is a local minimum of (B), by definition [9, Chapter 12], there exists a neighborhood $\mathcal{N}$ around y^* such that

$$g(y) \geq g(y^*) \quad \forall y \in \mathcal{N} \cap \mathcal{F}_B. \tag{33}$$

Define $\mathcal{S}_B = \mathcal{N} \cap \mathcal{F}_B$.

Since T is a diffeomorphism, it is in particular a homeomorphism. Therefore, T maps the open neighborhood $\mathcal{N}$ to an open set $T(\mathcal{N})$ which contains $T(y^*) = x^*$ [8, Chapter 12].

Define $\mathcal{M} = T(\mathcal{N})$. Let $\mathcal{S}_A = \mathcal{M} \cap \mathcal{F}_A$. Then,

$$\mathcal{S}_A = T(\mathcal{N}) \cap T(\mathcal{F}_B) = T(\mathcal{N} \cap \mathcal{F}_B) = T(\mathcal{S}_B) \tag{34}$$

where the second equality follows from the fact that T is in particular a bijection.

Therefore, for any $x \in \mathcal{S}_A$, there exists a unique pre-image $y = T^{-1}(x) \in \mathcal{S}_B$. From (33), we have:

$$\begin{aligned} g(y) &\geq g(y^*) \quad \forall y \in \mathcal{S}_B \\ \Rightarrow \quad f(T(y)) &\geq f(T(y^*)) \quad \forall y \in \mathcal{S}_B \\ \Rightarrow \quad f(x) &\geq f(x^*) \quad \forall x \in \mathcal{S}_A. \end{aligned} \tag{35}$$

We have shown the local optimality of x^* in (A). Since (A) is a convex optimization problem, any locally optimal solution is also globally optimal. Therefore, x^* is a global minimum of (A). □

Next, we show that this applies to our specific problem.

Proposition 2 ([4]). *The Cholesky map $\mathcal{L}$ is a diffeomorphism between smooth manifolds $\mathbb{L}_+^n$ and $\mathbb{S}_+^n$.*

The covariance steering problem without chance constraints has been shown to be a convex optimization problem [5]. Therefore, by Proposition 1, the locally optimal solution obtained from the SCP algorithm is globally optimal.

6 Numerical Examples

The convex optimization problems are solved in MATLAB using YALMIP [6] and MOSEK [7] on a MacBook Pro with an M4 Max processor with 24GB of RAM. We use the same SCP parameters as in [10].

Some practical implementation tips:

- Since the algorithm requires iteratively linearizing the problem, an initial guess is required for each of the variables. The initial guess for S_k can be obtained by linearly interpolating between the Cholesky factors of the initial and terminal covariances (There are several other options for the interpolation of positive definite matrices [4], but a comparison of these is beyond the scope of this paper). The initial guess for L_k can be set to zero.

- The SCP algorithm allows the user to choose whether to impose a trust region on the variables, to ensure that the solution remains in a region where the linear approximation is accurate. We observed that a trust region on L_k is not necessary, and causes chattering close to the solution. Trust regions on μ_k and v_k are not necessary, since they only appear in the convex constraints and are not subject to the linearization.

6.1 Comparison with Full Covariance Approach

We consider the numerical example in [5] with the following parameters:

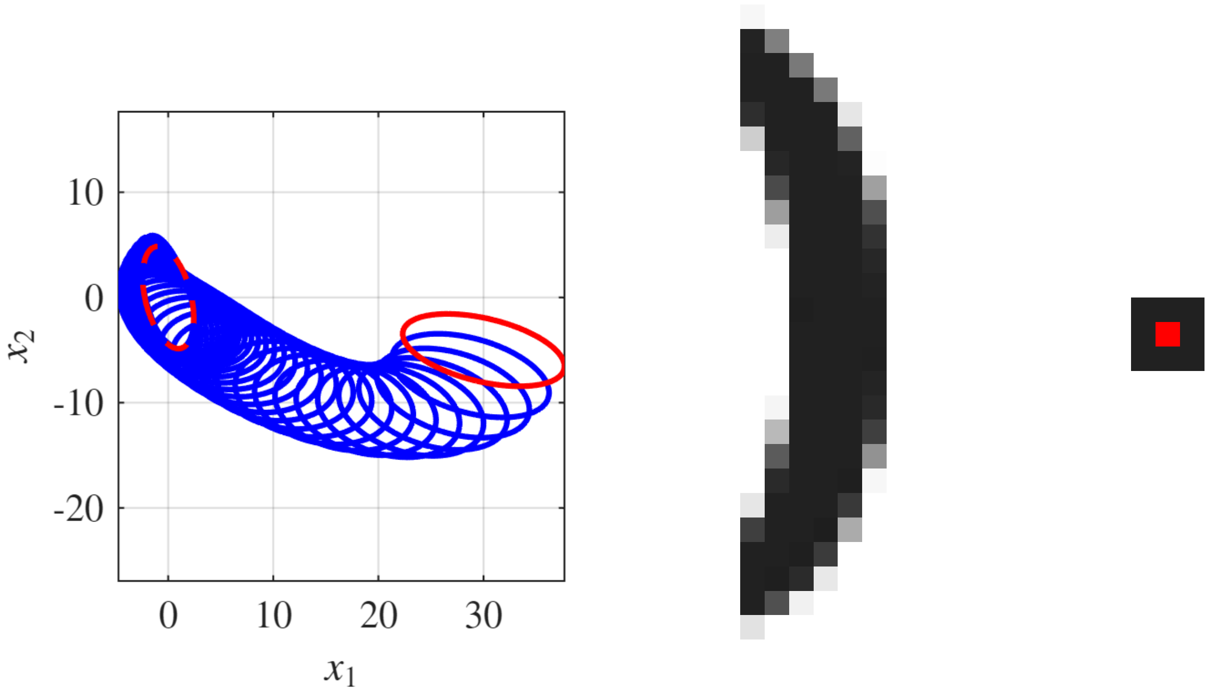
The cost function value match to 7 digits between the full covariance steering and the square root QR covariance steering methods.

6.2 Chance constraints

Showcase the chance constraints handling capability and compare solution times with full covariance and block approaches

7 Discussion

8 Conclusions



(a) Full covariance steering result

(b) Square root QR covariance steering result

Figure 1: Comparison of covariance steering results

Figure 2: Convergence history of the `SCvx*` algorithm for the proposed method

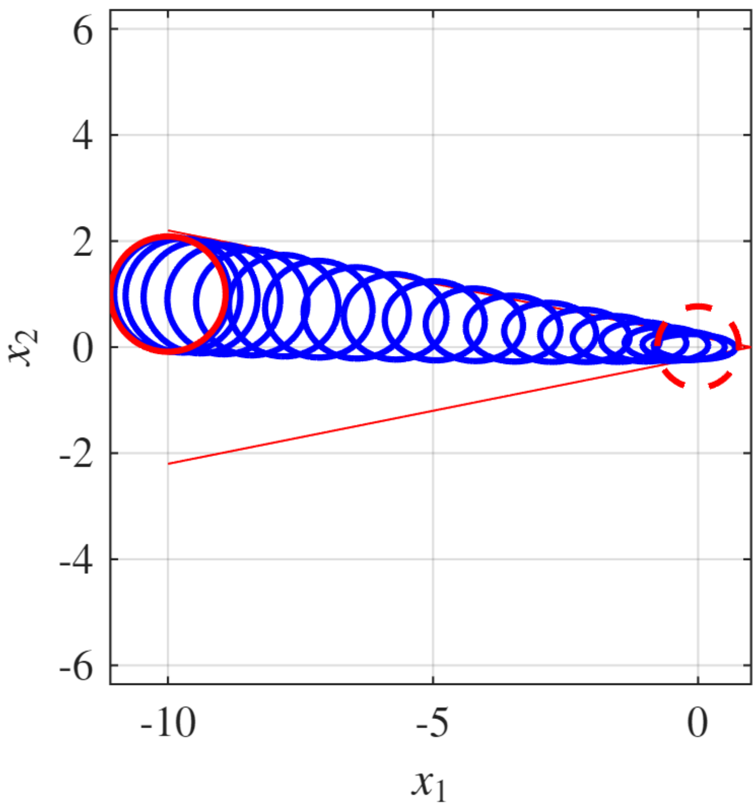


Figure 3: Example trajectory under chance constraints, solved using the proposed method

=== TERMINATION STATUS COMPARISON ===

Trial	Full Cov Status	SqrtQR Status	FullCovOptVal	SqrtQROptVal	Runtime (s)
1	SOLVED	SOLVED	2.73e+01	2.73e+01	0.95
2	NUMERICAL	STALLED	1.51e+07	N/A	3.33
3	SOLVED	SOLVED	5.77e+00	5.77e+00	0.78
4	SOLVED	SOLVED	7.41e+00	7.41e+00	0.76
5	SOLVED	SOLVED	1.17e+01	1.17e+01	0.62
6	SOLVED	SOLVED	1.56e+01	1.56e+01	0.77
7	SOLVED	SOLVED	1.20e+02	1.20e+02	0.97
8	SOLVED	SOLVED	9.66e+00	9.66e+00	0.72
9	NUMERICAL	STALLED	6.25e+04	N/A	3.00
10	NUMERICAL	STALLED	3.39e+05	N/A	2.15
11	SOLVED	SOLVED	1.21e+02	1.21e+02	1.36
12	SOLVED	SOLVED	1.33e+01	1.33e+01	0.65
13	SOLVED	SOLVED	6.16e+01	6.16e+01	1.07
14	SOLVED	SOLVED	6.53e+02	6.53e+02	3.58
15	SOLVED	SOLVED	1.21e+01	1.21e+01	0.65
16	SOLVED	SOLVED	1.62e+01	1.62e+01	0.72
17	SOLVED	SOLVED	2.11e+02	2.11e+02	1.65
18	SOLVED	SOLVED	1.58e+01	1.58e+01	0.71
19	SOLVED	SOLVED	4.02e+01	4.02e+01	0.66
20	SOLVED	SOLVED	1.85e+01	1.85e+01	1.07

Figure 4: Comparison of solutions times between full covariance and square root QR covariance steering methods

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